

Work in Progress: AI-Guided EGD Coverage Compliance via SSS-Kensho-Yahoo Protocol

Introduction

In recent years, there has been a significant increase in the application of artificial intelligence (AI) in gastroenterology, especially for improving diagnostic accuracy during esophagogastroduodenoscopy (EGD) procedures[1][2][3]. This study interested in the development of a real-time AI-based system that is intended to aid clinicians during EGD by providing help in visualization of the gastric mucosa, thus minimizing the chances of missing blind spots. Despite the significant advances in the development of AI for gastrointestinal endoscopy, one of the main limitations of current research is its dependence on simplified examination protocols, including the European Society of Gastrointestinal Endoscopy (ESGE) 4-image framework, the SAC (Systematic Alphanumeric Coding) system, and image-enhanced endoscopy (IEE) techniques [4].

While artificial intelligence methods like ENAD CAD-G have been found effective in assisting with lesion detection and compliance with simple protocols, such methodologies are usually incapable of thoroughly examining the stomach owing to their narrow focus and time limitations. To counter these limitations, this paper presents an AI algorithm trained on the SSS-Kensho-Yahoo protocol, a systematic and exhaustive approach that breaks down the stomach into 21 zones for the purposes of ensuring exhaustive visual examination. Unlike existing algorithms, the SSS-Kenshō protocol prioritizes the achievement of complete mucosal coverage, a technique that has been associated with increased detection and decreased missed abnormality rates [5]. The principal argument of this research is that applying artificial intelligence to the SSS-Kensho protocol has the potential to improve procedural quality, minimize diagnostic errors, and promote standardized clinical practice across a broad variety of healthcare settings, including inured environments beyond academia.

1. AI in Medicine

Artificial intelligence is becoming increasingly central to modern healthcare by improving diagnostic performance, tailoring treatment to the individual patient, reducing the administrative burden, and streamlining the drug development process. These technologies are having a deep impact on the interactions between patients and healthcare professionals in healthcare services [6][7]. In diagnostics, and specifically in medical imaging, AI—most notably deep learning algorithms—has been able to equal or even surpass medical experts in the evaluation of radiological images and dermatological diseases. This has enabled the earlier and more accurate detection of diseases such as pneumonia, melanoma, and diabetic eye disease [2][7]. For example, in diabetic retinopathy, AI systems have been shown to be highly accurate and reliable, offering an inexpensive and scalable solution to chronic disease [1].

In addition to imaging-based diagnostics, artificial intelligence is increasingly becoming a central tool in clinical decision-making. It utilizes vast electronic health records to assist healthcare professionals in developing personalized treatment plans, streamline patient documentation procedures, and optimize medication management. This capability not only minimizes the risk of human error but also allows clinicians to devote more time to interacting with patients [14][15]. Artificial intelligence is also transforming pharmaceutical research by assisting scientists in identifying potential drug candidates more quickly and

enhancing the design and architecture of clinical trials. These advances have the potential to lower research costs and accelerate the introduction of new therapies into the market [16].

Beyond the clinical environment, AI is automating hospital administration—scheduling, billing, and resource allocation—so that healthcare providers can have more time in direct patient contact. As the technologies continue to improve, they have the potential to make high-quality care more universally accessible, close gaps in access, and support a more preventive and continuous approach to health by examining real-time patient data and recommending early interventions. While issues of data bias, ethics, and transparency continue to afflict the technology, ongoing advances in AI research are increasingly pushing the boundaries of what is possible, paving the way for its central role in future healthcare systems[1].

2. AI in Gastroenterology

Artificial intelligence has rapidly emerged as a force to be reckoned with in gastroenterology, transforming the practice of making diagnoses, conducting procedures, and directing therapies. Its use in gastrointestinal endoscopy—specifically, for upper GI procedures like esophagogastroduodenoscopy (EGD)—has brought about significant gains in the detection of abnormalities, evaluation of lesions, and optimization of procedural efficiency [10][22]. Early innovations in the field centered on identifying tumors, forecasting lesion pathology, and detecting *Helicobacter pylori* infection. Such early systems were very fast and precise, with some artificial intelligence systems diagnosing *H. pylori* in a matter of seconds—lightyears beyond the speed and activity of expert clinicians [17][18].

Artificial intelligence-based computer-aided detection (CADE) and computer-aided diagnosis (CADx) systems used in EGD procedures are being designed to identify distinct anatomical sites, reduce missed areas while scanning, and establish consistency with the manner of procedures. Studies have revealed that such AI models are capable of categorizing over 97% of endoscopic images correctly and enhance the completeness of the procedure, with a level of close to 90% completeness realized in certain studies [14]. Furthermore, these algorithms have been extremely accurate in the differentiation of different submucosal tumor types, for instance, separating gastric mesenchymal from non-mesenchymal tumors correctly, and even performing better than experienced endoscopists. They have also demonstrated excellent ability in assessing conditions such as gastroesophageal varices and estimating the risk of bleeding—occasionally providing more accurate estimates than clinicians in risk stratification and recommending treatment regimens [8].

Clinical trials have repeatedly proven that AI systems reduce the chances of missing cancerous or precancerous lesions during endoscopy and hence facilitate the early detection of upper GI cancer and potentially improved patient outcomes [9]. The capability of AI to scan high-definition endoscopic images makes it capable of diagnosing slight neoplastic alterations at an early stage, predicting postoperative complications, and assisting more individualized therapy based on analysis of genetic information and medical history. Ever since 2019, the research work in the development of real-time AI platforms for GI endoscopy has moved ahead at a fast pace with huge importance placed on the development of better polyp detection, image segmentation, and classification. These technologies are one aspect of the larger movement in medicine towards precision medicine and more non-invasive methods of diagnosis [11].

3. AI in EGD

Artificial intelligence has introduced significant improvements in esophagogastroduodenoscopy (EGD), an essential diagnostic gastroenterology procedure, by

enhancing the detection of abnormalities in the stomach, improving procedural consistency, and providing assistance to less-experienced professionals [9]. Of particular interest is AI-powered system such as the EW10-EG01 [11], which has demonstrated a high percentage improvement in tumor detection rates during EGD. Research shows that employing AI can almost double the proportion of detecting esophageal and gastric tumors—0.78% in the case of AI versus 0.37% with conventional methods [11]. Such computer vision systems scan real-time endoscopic video feeds and highlight suspicious regions, assisting doctors in spotting flat or subtle lesions that are usually missed upon standard screening for gastric cancer. In addition, AI technologies help ensure full visual coverage of the stomach by giving real-time feedback on which regions have been scanned and which are still unchecked, thus minimizing the chances of blind spots during the procedure. Apart from the detection of lesions, AI has proved to perform robustly in more specific diagnostic uses, including the detection of *Helicobacter pylori* infection at accuracy rates of 93% to 97%, assessment of gastroesophageal varices, and determination of the depth of invasion in early gastric cancer at more than 90% accuracy [12]. Although these are encouraging results, the use of AI in standard EGD procedures has been associated with increased examination times and a higher number of biopsies, since more abnormalities are identified for further assessment.

4. Different EGD Protocols

The procedure for performing a proper visual examination of the stomach during esophagogastroduodenoscopy (EGD) has been significantly improved by the evolution of systematic protocols to enhance diagnostic accuracy, reduce missed lesions, and enhance image documentation consistency (Role of Artificial Intelligence-Guided Esophagogastroduodenoscopy in Assessing the Procedural Completeness and Quality). Two of the most well-established methods in this field are the Systematic Screening Protocol for the Stomach (SSS), commonly known as the Kenshi Yao method, and the Systematic Alphanumeric Coded Endoscopy (SACE) system.

The SSS protocol, as formulated by Dr. Kenshi Yao, mandates the acquisition of 22 unique, non-overlapping images that cover all the anatomical areas of the stomach—i.e., the antrum, corpus, and fundus[5]. This is done using quadrant-based imaging, in addition to the employment of pharmacological agents to inhibit peristalsis and the employment of air insufflation to maximize mucosal exposure by removing folds and mucus[4][12]. This organized mapping of the gastric mucosa has been linked with enhanced detection of clinically relevant lesions and increased procedural completeness. For example, reports cite an increase in lesion detection rates (5.45% vs. 3.48%) and increased examination completeness to 89.2% when combined with AI-powered anatomical identification software [12].

Conversely, the SACE protocol utilizes a serial documentation system with 21 overlapping image zones. It labels each region using a coded alphanumeric system, supporting standardization, clarity, and simpler comparison of various sections when cross-referencing [4]. Although it possesses similar systematic design to the SSS protocol, SACE is differentiated by its overlapping imaging technique and incorporation into complete evaluations of the upper GI tract beyond the stomach.

The other widely used method is the European Society of Gastrointestinal Endoscopy (ESGE) protocol, which supports a rapid four-image acquisition—of the antrum, angulus,

body, and fundus. While this method is supposed to be quick, its narrow focus is criticized for the risk of undertated findings versus more labor-intensive protocols [13].

5. Debate on the Protocols

Present debate regarding efficacies of various endoscopic procedures protocols revolves mainly around the balance between diagnostic completeness and clinical convenience and procedure time. The SSS-Kenshō protocol provides the most amount of detail to be available, which offers full imaging of the gastric mucosa along with highly enhanced detection capacity of lesions. However, this benefit comes at the cost of longer examination durations—approximately 5.3 minutes longer than unstructured approaches—and requires training to ensure homogeneity in its application by practitioners [4][5]. On the other hand, the ESGE 4-image strategy is used in high-volume clinical practice due to it being easy and effective but has been criticized for undercoverage, potentially omitting important anatomical areas and thus increasing the risk of missed diagnosis [13].

The SACE protocol attempts to compensate for this shortcoming by the use of overlapping image zones to reduce blind spots. For all its sequential organization, incorporation of complex coding by SACE and redundancy of imaging has ensured that it did not achieve widespread acceptance. Criticizers would argue that conventions of naming and overlap do not necessarily translate into improved diagnostic results over the likes of SSS, whose agenda is anatomical comprehensiveness and crisp visualizations

The protocol selected may profoundly affect accuracy in diagnosis, especially in the detection of early-stage or subtle gastric cancers easily overlooked during less thorough tests. There is mounting evidence to suggest the benefits of systematic and detailed protocols. Studies have revealed that up to 11.3% of gastric cancers can be missed during endoscopies, with inadequate mucosal examination being cited as an important contributing factor[19]. Therefore, while less efficient, guidelines like SSS can ideally have the best feasibility-diagnostic sensitivity balance—especially when augmented with artificial intelligence algorithms that facilitate compliance with protocols and real-time process quality monitoring [5][13].

6. AI in Non SSS Protocols

Advances in the area of artificial intelligence have paved the way for advanced models trained directly on formally designed stomach examination protocols, providing good improvements to diagnostic precision and procedural uniformity when conducting esophagogastroduodenoscopy (EGD)[20]. One such example is the ENAD CAD-G system, which uses the Systematic Alphanumeric Coding (SAC) system protocol. This AI system supports computerized detection of lesions, with abnormalities allocated to defined segments of anatomy ranging from B3 in the antrum through to F2 in the fundus, an improvement of 42% for the detection of early gastric cancer (EGC) over non-AI traditional methods [13]. Likewise, the ENDOANGEL-ME system employs deep learning techniques using the magnifying image-enhanced endoscopy (M-IEE) to perform EGC achieving clinical trial results with 92.9% sensitivity and 91.7% specificity. ENDOANGEL-ME is one more real-time AI system developed for early gastric cancer (EGC) diagnosis using magnifying image-enhanced endoscopy (M-IEE). It employs convolutional neural networks (CNNs), a type of deep learning architecture effective in image analysis. Trained on over 4,600 images from six hospitals and validated on multiple datasets and real clinical videos, ENDOANGEL-ME achieved diagnostic accuracies exceeding 88% and outperformed experienced endoscopists. In real-world settings, it demonstrated a sensitivity of 92.59% and improved clinicians'

diagnostic performance [21].

Despite these advancements, several challenges remain in the path of broader clinical uptake. One of these is increase in the examination time, with AI systems adding about 36 seconds per procedure—a noteworthy factor in high-volume settings. Additionally, most AI algorithms are restricted to visual input alone, without incorporating patient-specific clinical factors like medical history, medication lists, or tailored risk factors. One more major drawback is low generalizability, most systems have experienced a significant decline in diagnostic skill when brought to clinical settings[21]. In addition, most AI systems are trained and validated on individual datasets—largely Chinese-based—restricting ethnic and procedural variation of training sets and excluding external validation. Clinical application is also clouded by interinstitutional variability in image quality, interobserver variability in annotation, and a tendency towards false positives due to intrinsic mucosal folds or suboptimal gastric inflation. Finally, current AI technology is not likely to possess the capability to dynamically track lesions on live video streams, which restricts their application in real-time, continuous evaluation during procedures [21].

7. AI in the SSS Kenshi Yao Protocol

One of the breakthroughs in egd is the WISENSE system [18][8], which applies a combination of deep convolutional neural networks (CNNs) and deep reinforcement learning (DRL) to assist anatomical site identification during esophagogastroduodenoscopy (EGD) after sss protocol. This framework has been thoroughly trained on a broad dataset containing controlled laboratory images, actual-patient information, and clinical cases, enabling it to split 26 various regions of the stomach anatomy with high spatial precision.

The main function of WISENSE is to give real-time guidance on whether all the major areas of the stomach have been sufficiently imaged during the procedure. This is achieved via frame-level landmark identification in line with the emphasis of the SSS protocol on the extensive documentation of specific sites. The WISENSE device is particularly useful in differentiating anatomically proximate yet diagnostically different regions—such as the greater curvature of the antrum and the anterior wall of the gastric body—often not seen with conventional endoscopy[18]. In helping to reduce areas of unexamined tissue and maximize image acquisition, WISENSE is of substantial benefit, especially in clinical environments when differences in practitioner skill affect the quality of the examination [8].

Yet, the system's present limitations are evident. Its primary focus is spatial recognition, and it does not account for time-based progression of an endoscopic procedure. Since extensive gastric inspection not only entails visualization of specific areas but also their systematic and ordered examination, the lack of temporal modelling in WISENSE limits its procedural knowledge. For instance, it cannot determine whether the migration from one zone to another was along a rational course of navigation or whether there were areas that were redundantly surveyed and some omitted.

Furthermore, WISENSE does not have the capability to track interruptions in the sequence or warn practitioners of possible deviations from procedure in real-time. That is, while it can retrospectively confirm that regions of anatomy have been covered, it cannot be employed as an active guide that controls the workflow of the examination. Its ability to provide assurance of compliance with the SSS's sequential order is therefore limited, and its role is mostly confined to spatial verification. In the future, incorporating such systems real-time procedural monitoring and temporal context will be key to unlocking the full potential of AI in endoscopic diagnosis.

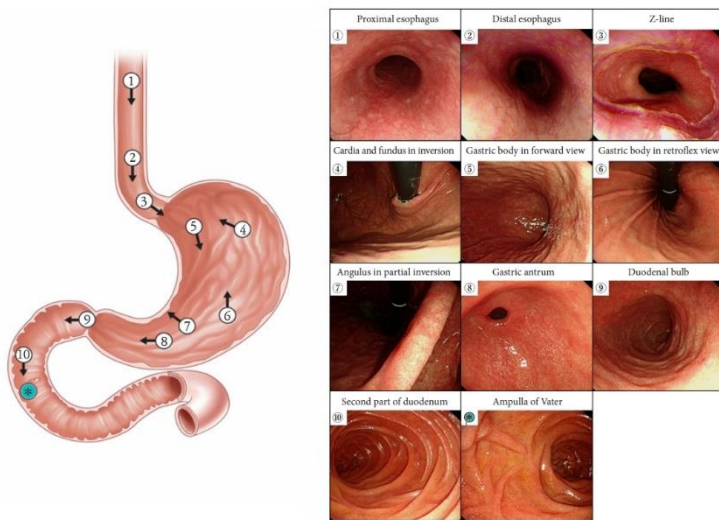
8. The Gap in Real-Time Procedural Guidance

While devices such as WISENSE are a major step forward in structured gastric endoscopy, an inherent limitation remains—present-day AI technologies are not yet advanced enough to control the procedural flow of an esophagogastroduodenoscopy (EGD) in real time. The Kenshi Yao SSS protocol was designed specifically to deal with repeated problems associated with incomplete mucosal examinations. It describes a systematic technique that involves at least 22 well-defined anatomical locations, with emphasis on quadrant-based imaging of the stomach's principal areas: the antrum, body, and fundus. It also underscores the significance of well-performed preparation steps like the attainment of sufficient luminal distension, mucus removal, and peristalsis suppression for optimal visualization conditions. Clinical evidence is overwhelmingly supportive of the SSS approach efficacy; in 2020, a study reviewing more than 10,000 EGD cases revealed that adherence to protocol substantially enhanced rates of lesion detection (5.45% vs. 3.48%) and the number of images per case documented (32.7 vs. 19.9) [5].

Despite such robust data, current AI systems do not fully incorporate the sequential decision-making framework inherent in the SSS. Leading models like ENDOANGEL, ENAD CAD-G, and WISENSE primarily rely on convolutional neural networks (CNNs) to extract spatial features. While some models offer anatomical labeling or region classification, they typically do not track the temporal progression of the examination. Furthermore, there are no existing AI platforms that can evaluate whether the procedural sequence was followed in a logical and comprehensive manner[18]. As yet, no built-in tools are available to diagnose whether or not a practitioner omitted critical sections or strayed from the anticipated anatomical course. For instance, if an endoscopist moves from the fundus to the antrum without a complete examination of the body of the stomach, current AI would not identify or warn the operator of this omission. Similarly, such systems fail to offer navigational recommendations or reminders to re-examine omitted areas, losing the chance to be real-time procedural assistants.

This is not a technical limitation only; it's a wider conceptual deficit in the way AI is being used in endoscopy. The vast majority of systems view the procedure as a series of discrete images instead of a continuous, structured process that progresses over time. The absence of this temporal dimension significantly hinders AI's ability to function as a procedural partner—ensuring not only that abnormalities are found but that the entire examination is conducted systematically and thoroughly. Overcoming this challenge will require a new generation of AI models that integrate both spatial recognition and temporal logic. This integration can be done by using complex machine learning frameworks such as recurrent neural networks (RNNs), temporal convolutional networks (TCNs), or models based on the Transformer. Only by infusing these features can AI transition from being a diagnostic tool to an actual guide along the endoscopic process.

Fig: Different regions of stomach



Our Dataset:

The recently released GastroHUN dataset is a big leap in gastric endoscopy research by providing a comprehensive, high-quality resource after a structured photographic documentation procedure. With 8,834 annotated images and 4,729 labeled video sequences from 387 patients, all captured with the Systematic Screening of the Stomach (SSS)

Kenshi Yao protocol, this dataset is unique because of its clinical correctness and its usability for research. It carefully captures 22 stomach anatomical landmarks, including the antrum, body, fundus, and incisura, by anterograde and retroflex views to offer full mucosal examination and minimize blind spots. The dataset further includes an additional "NA" class to include unqualified images, again making it stronger by handling real-world variability[23].

One of the fundamental strengths of GastroHUN lies in the expert annotations. All video sequences and frames were annotated by four trained reviewers that comprised both junior and senior gastroenterologists, ensuring consistency and accuracy at the expert level. To establish reliability validation of these annotations, Cohen's kappa coefficient was employed for the verification of inter- and intra-rater agreement with pairwise cautionary comparisons and repeated-image testing, thus giving clear evidence of label reliability and eliminating bias.

Furthermore, the data is organized according to a stratified partitioning scheme—70% training, 15% validation, and 15% testing—that ensures balanced representation of labels with high inter-rater agreement levels across subsets using Fleiss' kappa-based stratification. This makes GastroHUN highly appropriate for AI algorithm benchmarking in uniform and realistic test environments. The dataset is interesting in the sense that it presents both image-level and sequence-level annotations, from which one may train and validate AI models for processing static images as well as dynamic video sequences.

By delivering a strict, high-yield clinical protocol (SSS), exact and formalized annotations from multiple experienced assessors, and supporting multi-modal training (image and video), GastroHUN creates a unique and established platform for gastric cancer detection AI research, anatomical localisation, quality assurance, and endoscopic protocol conformity. It completes an essential public dataset gap, especially targeting the stomach region, an area that has not been as fully represented in open-source repositories. With its clinical accuracy, dataset density, expert validation, and methodological transparency, GastroHUN sets a new benchmark for completeness, annotation quality, and research potential in gastrointestinal imaging, as a highly valuable resource for developing AI models within this area[24].

Methodology:

To address the main limitation of existing AI-based endoscopy systems—their inability to understand the procedural flow during EGD—we created a multi-phase approach that is able

to capture both spatial and temporal dynamics in video-based gastric examination. In accordance with the Systematic Screening of the Stomach (SSS) Kenshi Yao protocol, which describes a 22-point anatomical examination plan using quadrant-based imaging and preparatory steps, our approach mimics not only anatomical features but also procedural sequence.

We began by training a Vision Transformer (ViT) model on supervised still images of endoscopic procedures to identify high-level spatial features. Concurrently, we trained a ResNet CNN convolutional neural network and a ConvNeXt-Tiny model to act as CNN-based baselines for spatial classification purposes to enable comparative performance evaluation.

To address the temporal aspect, we trained a Video Vision Transformer (ViViT) on labeled video segments that included a center frame and six side frames to see short-term patterns. Although ViViT was promising, computational resources limited its scalability. To capture longer-term temporal dependencies, we investigated recurrent neural network (RNN)-based models and sequence-to-sequence frameworks.

Our best architecture so far is a hybrid CNN-LSTM pipeline, which leverages convolutional feature extraction with sequential modeling through LSTM layers, finding a balance between temporal understanding and computational cost appropriate for real-time use.

Heavy data preprocessing was used, such as normalization, random jitter, flipping, rotation, and sequence alignment. Video samples were synchronized with annotated frames carefully and padded with neighboring frames to maintain temporal transitions during training.

Results and Evaluation:

Fig: Resnet50 Training

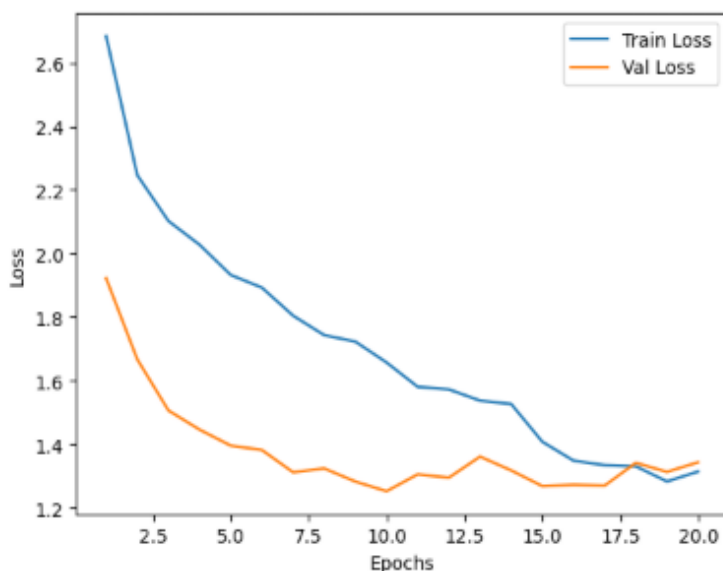
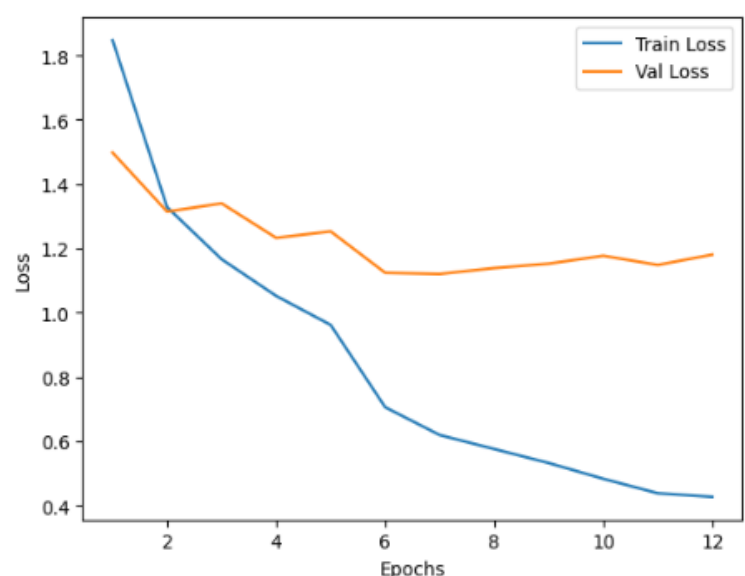


Fig: ConvNeXt-Tiny Model



Our ViT-based image classification model proved effective for static anatomical classification, achieving a macro F1-score of 67%, confirming its capacity to correctly identify anatomical regions in accordance with the SSS Kenshi Yao protocol.

The ResNet model also delivered competitive performance on the test set of 1,353 images:

- Accuracy: 66.15%
- Macro Average: Precision = 0.6625, Recall = 0.6626, F1-score = 0.6608
- Weighted Average: Precision = 0.6626, Recall = 0.6615, F1-score = 0.6602

We also trained a ConvNeXt-Tiny model to assess the performance of a modern lightweight CNN. On the same test set, it achieved:

- Accuracy: 61.42%
- Macro Average F1-score: 0.6112
- Weighted Average F1-score: 0.6099

Below is a breakdown from the ConvNeXt-Tiny classification report :

- Strongest performance was seen in P6 (F1 = 0.8033), L6 (F1 = 0.7963), and G4 (F1 = 0.7350).
- Moderate to strong recall was observed across regions like G1 (0.7778), G2 (0.7324), and A4 (0.7581).
- The model struggled with classes such as L3 (F1 = 0.5000), L5 (F1 = 0.4444), and P4 (F1 = 0.5275), suggesting room for improvement in visual discriminability for those sites.

Early experiments with RNNs for long sequence modeling showed overfitting, with training-validation accuracy gaps despite regularization. Validation accuracy rose from 13.6% to 17.2%, then plateaued, triggering early stopping. Despite this, RNNs retained some temporal memory and showed potential with further tuning.

ViViT, though conceptually promising for video-based learning, was constrained by high memory demands. These experiments underlined the importance of computational efficiency in deploying large-scale temporal models in real-world endoscopic systems.

Our most promising direction so far is the CNN-LSTM hybrid model, combining robust CNN-based feature extraction with temporal modeling via LSTMs. Preliminary results indicate it offers improved real-time procedural alignment, supporting accurate and continuous tracking of anatomical progress in EGD.

Conclusion:

Our work marks a structured and progressive effort to address the longstanding challenge in AI-assisted endoscopy—the lack of awareness of procedural flow—by introducing temporal modeling into systems traditionally focused on individual frames. Unlike existing solutions such as WISENSE and ENDOANGEL, which concentrate primarily on spatial classification, our approach takes a step forward by incorporating temporal logic inspired by the SSS protocol. Model development and evaluation of models like ViT, ViViT, and RNN-based models, and more recently, the addition of a CNN-LSTM hybrid, indicate a thorough investigation of this problem. Although there are still challenges, especially with regards to obtaining uniform video-level accuracy on long sequences, the advancement so far demonstrates the viability and feasibility of adding temporal consciousness to AI and improving procedural completeness and lessening the likelihood of missing lesions. We are further investigating novel architectures, including temporal attention models and Transformer-based architectures, to further enhance real-time guidance. This research opens the door to an intelligent endoscopy assistant that not only identifies anatomical landmarks

but also verifies that the examination adheres to gold-standard protocols, preserving procedural integrity.

References:

1. Xu, N., Yang, D., Arikawa, K., & Bai, C. (2023). Application of artificial intelligence in modern medicine. *Clinical eHealth*, 6, 130–137. <https://doi.org/10.1016/j.ceh.2023.09.001>
2. Buch, V. H., Ahmed, I., & Maruthappu, M. (2018). Artificial intelligence in medicine: Current trends and future possibilities. *The British Journal of General Practice*, 68(668), 143–144. <https://doi.org/10.3399/bjgp18X695213>
3. Lazăr, D. C., Avram, M. F., Faur, A. C., Romoșan, I., & Goldiș, A. (2021). The role of computer-assisted systems for upper-endoscopy quality monitoring and assessment of gastric lesions. *Gastroenterology Report*, 9(3), 185–204. <https://doi.org/10.1093/gastro/goab008>
4. Machaca Quea, N. R., Emura, F., Barreda Bolaños, F., Salvador Arias, Y., Arévalo Suárez, F. A., & Piscoya Rivera, A. (2016). Effectiveness of systematic alphanumeric coded endoscopy for diagnosis of gastric intraepithelial neoplasia in a low socioeconomic population. *Endoscopy International Open*, 4(10), E1083–E1089. <https://doi.org/10.1055/s-0042-115408>
5. Yao, K. (2013). The endoscopic diagnosis of early gastric cancer. *Annals of Gastroenterology*, 26(1), 11–22.
6. Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). AI in health and medicine. *Nature Medicine*, 28(1), 31–38. <https://doi.org/10.1038/s41591-021-01614-0>
7. Bajwa, J., Munir, U., Nori, A., & Williams, B. (2021). Artificial intelligence in healthcare: Transforming the practice of medicine. *Future Healthcare Journal*, 8(2), e188–e194. <https://doi.org/10.7861/fhj.2021-0095>
8. Wu, L., Zhang, J., Zhou, W., An, P., Shen, L., Liu, J., Jiang, X., Huang, X., Mu, G., Wan, X., Lv, X., Gao, J., Cui, N., Hu, S., Chen, Y., Hu, X., Li, J., Chen, D., Gong, D., ... Yu, H. G. (2019). Randomised controlled trial of WISENSE, a real-time quality improving system for monitoring blind spots during esophagogastroduodenoscopy. *Gut*, 68(12), 2161–2169. <https://doi.org/10.1136/gutjnl-2018-317366>
9. Dao, H. V., Nguyen, B. P., Nguyen, T. T., Lam, H. N., Nguyen, T. T. H., Dang, T. T., Hoang, L. B., Le, H. Q., & Dao, L. V. (2024). Application of artificial intelligence in gastrointestinal endoscopy in Vietnam: A narrative review. *Therapeutic Advances in Gastrointestinal Endoscopy*, 17, 26317745241306562. <https://doi.org/10.1177/26317745241306562>

10. Peng, Z., Wang, X., Li, J., Sun, J., Wang, Y., Li, Y., Li, W., Zhang, S., Wang, X., & Pei, Z. (2024). Comparative bibliometric analysis of artificial intelligence-assisted polyp diagnosis and AI-assisted digestive endoscopy: Trends and growth in AI gastroenterology (2003–2023). *Frontiers in Medicine*, 11.
<https://doi.org/10.3389/fmed.2024.1438979>
11. Adachi, K., Ebisutani, Y., Matsubara, Y., Okimoto, E., Ishimura, N., & Ishihara, S. (2025). Effectiveness of Artificial Intelligence in Screening Esophagogastroduodenoscopy. *Cureus*, 17(3), e79935.
<https://doi.org/10.7759/cureus.79935>
12. Lee, J. I., Kim, J. S., Kim, B.-W., & Huh, C. W. (2020). Taking More Gastroscopy Images Increases the Detection Rate of Clinically Significant Gastric Lesions: Validation of a Systematic Screening Protocol for the Stomach. *The Korean Journal of Helicobacter and Upper Gastrointestinal Research*, 20(3), 225–232.
<https://doi.org/10.7704/kjhugr.2020.0013>
13. Chang, Y. H., Shin, C. M., Lee, H. D., Park, J., Jeon, J., Cho, S.-J., Kang, S. J., Chung, J.-Y., Jun, Y. K., Choi, Y., Yoon, H., Park, Y. S., Kim, N., & Lee, D. H. (2024). Real-World Application of Artificial Intelligence for Detecting Pathologic Gastric Atypia and Neoplastic Lesions. *Journal of Gastric Cancer*, 24(3), 327–340.
<https://doi.org/10.5230/jgc.2024.24.e28>
14. Chen, Y., & Esmaeilzadeh, P. (2024). Generative AI in Medical Practice: In-Depth Exploration of Privacy and Security Challenges. *Journal of Medical Internet Research*, 26(1), e53008. <https://doi.org/10.2196/53008>
15. Noorbakhsh-Sabet, N., Zand, R., Zhang, Y., & Abedi, V. (2019). Artificial Intelligence Transforms the Future of Health Care. *The American Journal of Medicine*, 132(7), 795–801. <https://doi.org/10.1016/j.amjmed.2019.01.017>
16. Xu, M., Zhou, W., Wu, L., Zhang, J., Wang, J., Mu, G., Huang, X., Li, Y., Yuan, J., Zeng, Z., Wang, Y., Huang, L., Liu, J., & Yu, H. (2021). Artificial intelligence in the diagnosis of gastric precancerous conditions by image-enhanced endoscopy: A multicenter, diagnostic study (with video). *Gastrointestinal Endoscopy*, 94(3), 540-548.e4. <https://doi.org/10.1016/j.gie.2021.03.013>
17. Kröner, P. T., Engels, M. M., Glicksberg, B. S., Johnson, K. W., Mzaik, O., van Hooft, J. E., Wallace, M. B., El-Serag, H. B., & Krittanawong, C. (2021). Artificial

intelligence in gastroenterology: A state-of-the-art review. *World Journal of Gastroenterology*, 27(40), 6794–6824. <https://doi.org/10.3748/wjg.v27.i40.6794>

18. Abadir, A. P., Ali, M. F., Karnes, W., & Samarasena, J. B. (2020). Artificial Intelligence in Gastrointestinal Endoscopy. *Clinical Endoscopy*, 53(2), 132–141. <https://doi.org/10.5946/ce.2020.038>
19. Missing rate for gastric cancer during upper gastrointestinal endoscopy: A systematic review and meta-analysis | Request PDF. (2025). *ResearchGate*.
<https://doi.org/10.1097/MEG.0000000000000657>
20. Goenka, M. K., Afzalpurkar, S., Jejuri, S., Rode, G. A., & Tiwari, A. (2023). Role of artificial intelligence-guided esophagogastroduodenoscopy in assessing the procedural completeness and quality. *Indian Journal of Gastroenterology: Official Journal of the Indian Society of Gastroenterology*, 42(1), 128–135. <https://doi.org/10.1007/s12664-022-01294-9>
21. He, X., Wu, L., Dong, Z., Gong, D., Jiang, X., Zhang, H., Ai, Y., Tong, Q., Lv, P., Lu, B., Wu, Q., Yuan, J., Xu, M., & Yu, H. (2022). Real-time use of artificial intelligence for diagnosing early gastric cancer by magnifying image-enhanced endoscopy: A multicenter diagnostic study (with videos). *Gastrointestinal Endoscopy*, 95(4), 671–678.e4. <https://doi.org/10.1016/j.gie.2021.11.040>
22. Abadir, A. P., Ali, M. F., Karnes, W., & Samarasena, J. B. (2020). Artificial Intelligence in Gastrointestinal Endoscopy. *Clinical Endoscopy*, 53(2), 132–141. <https://doi.org/10.5946/ce.2020.038>
23. *GastroHUN an Endoscopy Dataset of Complete Systematic Screening Protocol for the Stomach* | *Scientific Data*. (n.d.). Retrieved May 3, 2025, from <https://www.nature.com/articles/s41597-025-04401-5>
24. Role of artificial intelligence-guided esophagogastroduodenoscopy in assessing the procedural completeness and quality | Request PDF. (n.d.). Retrieved May 2, 2025, from https://www.researchgate.net/publication/367559163_Role_of_artificial_intelligence-guided_esophagogastroduodenoscopy_in_assessing_the_procedural_completeness_and_quality

